INORGANIC CHEMISTRY CUMULATIVE EXAM

March 29, 2013

Almost all of the metals of the periodic table have been reacted with 2,4-pentanedione to form stable complexes. The chemistry of these complexes have been investigated by chemists with interests ranging from analytical to theoretical.

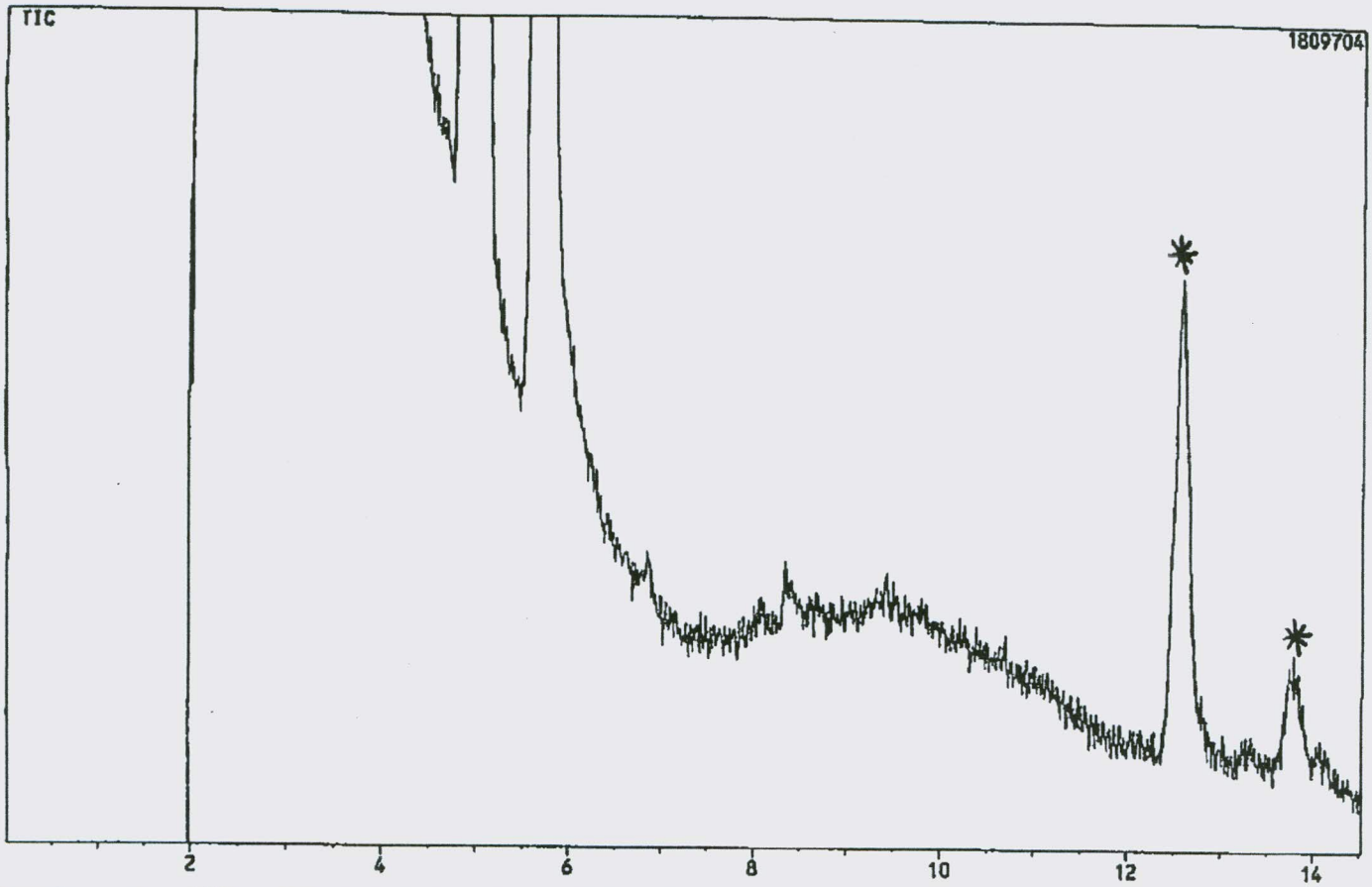
1. (40 points) A mixture of chromium nitrate and 1,1,1-trifluoro-2,4-pentandione was heated in ethanol to boiling for two minutes. The reaction mixture turned from blue to green. A small sample of the reaction mixture was subjected to analysis by GCMS. Two chromium containing peaks appeared in the GC trace (marked with *). The mass spectra (including the 511 parent ions) of the two GC peaks were nearly identical. The GC trace and mass spectrum are shown below. From this data describe the products and explain why the data leads to your answer. Write an equation that describes the overall reaction.

2. (20 points) If an analogous set of reactions is performed with 2,4-pentandione, 1,1,1-trifluoro-2,4-pentanedioneand 1,1,1,5,5,5-hexafluoropentanedione analogous products are obtained. However, the reaction conditions required were different for the three reactions. 30 minutes at reflux was required for 2,4-pentanedione. The hexafluoropentanedione reacts on contact. The reaction conditions for the trifluoropentanedione are described above. Explain the difference in reactivity.

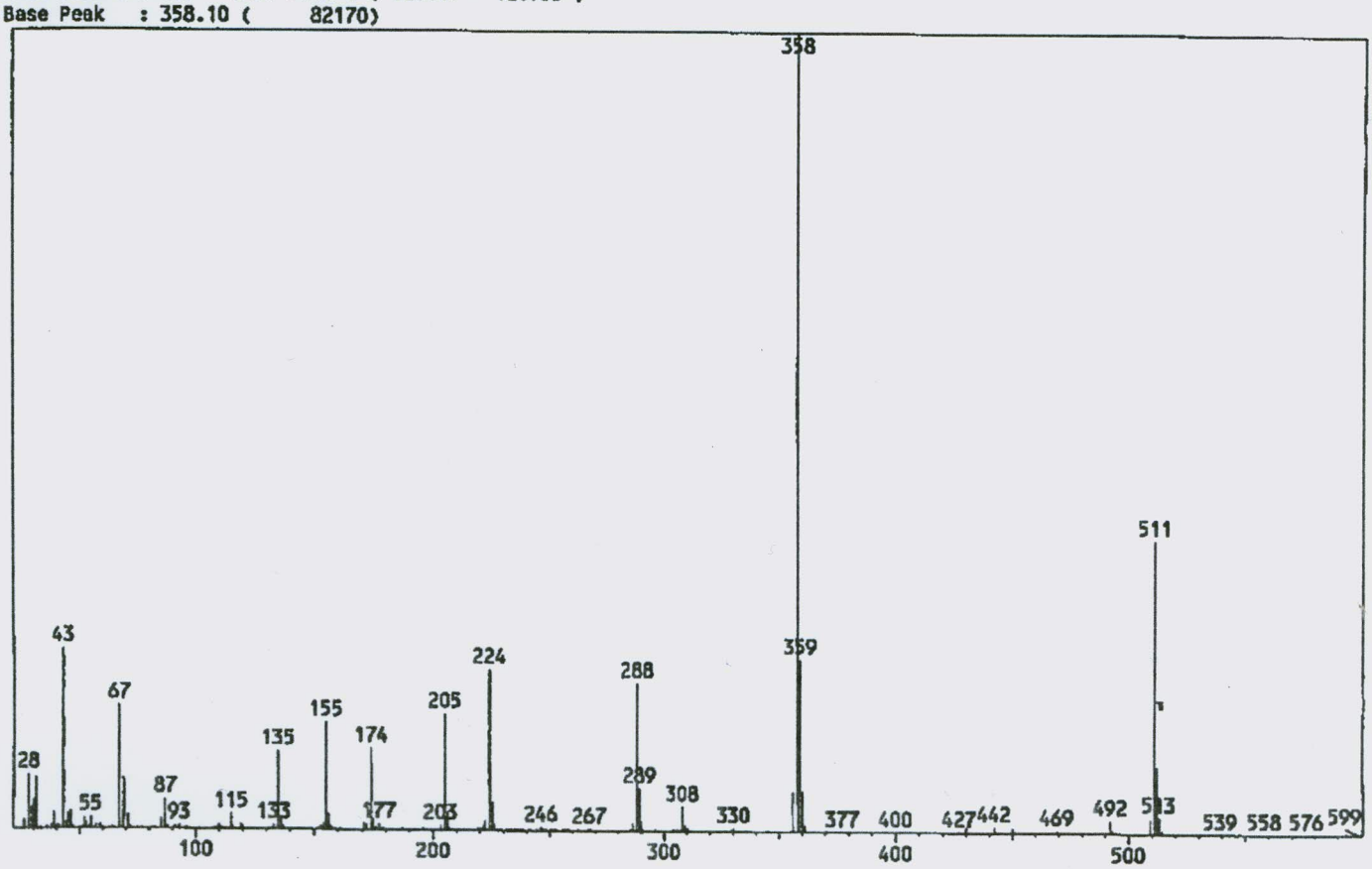
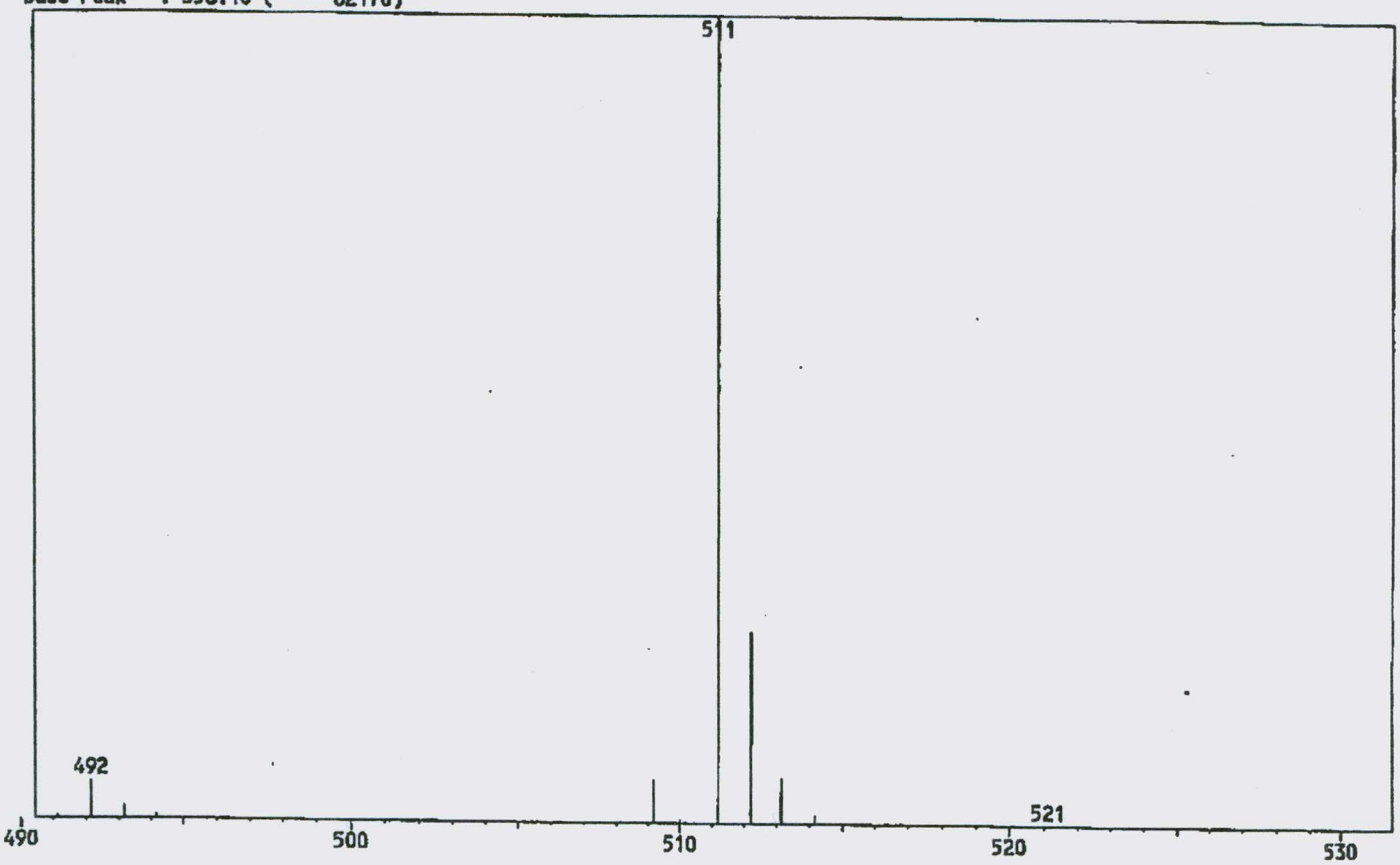
3. (20 points) In cases where reactivity is difficult, addition of the 2,4-pentanedione in the form of 2,4-pentanedionotothallium(I) to the metal chloride has proven effective. Suggest a structure for the thallium salt. Explain why this route might be superior to the reactions described above.

4. (20) points) From the details given above it should be evident that many of the complexes of 2,4-pentanedione are extremely stable. On the other hand there are no examples of complexes in which the coordination is satisfied by acetone anions. What features of pentandione make its complexes particularly stable in contrast to acetone anions?

DI Test



Scan # : (1490 - 1528) B.G. Scan # : (1541 - 1604)
Mass Peak # : 400 Ret. Time : (12.417 - 12.733)
Base Peak : 358.10 (82170)



Scan # : (1490 - 1528) B.G. Scan # : (1541 - 1604)
Mass Peak # : 400 Ret. Time : (12.417 - 12.733)
Base Peak : 358.10 (82170)